FA23

Dynamic Branch Prediction with Perceptrons

@nebu

Outline

Overview of Branch Prediction

Overview of Perceptrons and Related Math

The Paper

Section 1

Overview of Branch Prediction

What is a program counter?

- CPUs (logically) work by executing instructions one by one.
- The program counter points to the next instruction that the processor should execute.
- Typically, the PC just increments sequentially.
- Except…when?

What is a branch?

- Consider:

```
if (x == 2) {/* Do something */}
else {/* Do something else */}
```

- This compiles down to an instruction called a “conditional branch”, which changes the PC to some other value.
- The PC “jumps” to some other instruction.

Pipelining Requires *Prediction*

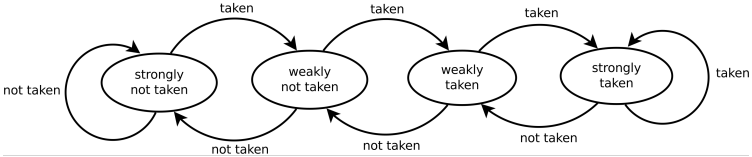
- Imagine a conditional branch instruction enters the pipeline.
- Until we figure out whether to branch (this requires going through a couple of pipeline stages…) what do we do?
- We can stall, but that's wasteful.
- Better to "predict" which way the branch will go to figure out where to issue next instructions from.

Ways to Predict

- "Static": always predict branches are taken (or untaken). Or some other static scheme, like always take backward branches, never forward branches, etc. etc.
- "Random": flip a coin to decide whether to take the branch. Makes timing nondeterministic.
- "Dynamic": much better, learns with the program at runtime.

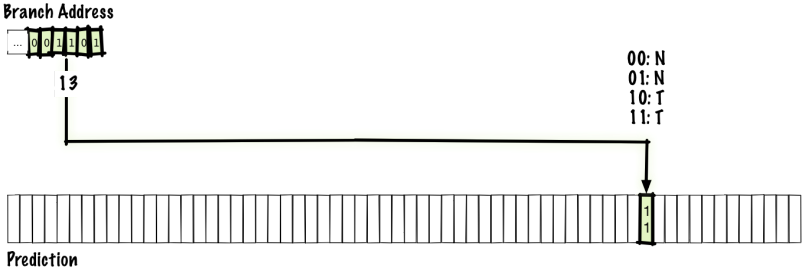
Dynamic Methods: One Level Saturating Counters

- One level, one bit: record the last outcome of the branch, assume the next one will go the same way.
- One level, two bit saturating counter.



- Better since *two* branches that go the other way are required to change state.

One-Level Indexing



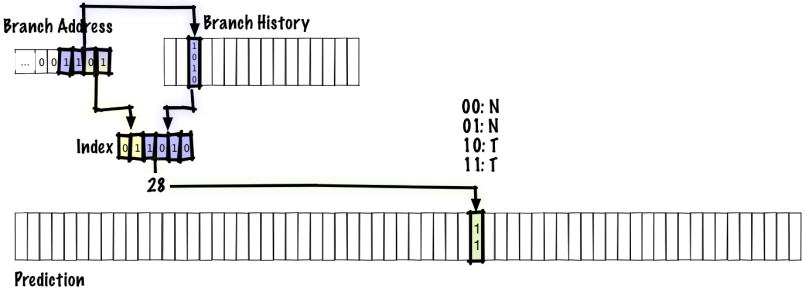
- There's a table of predictions indexed by the address of the instruction for predicting each branch.[1]

[1]Image credit: `https://danluu.com`

Dynamic Methods: Two Level Local

- Key idea is to catch repeating patterns of branches, for example: `100100100100`...
- Requires remembering the past n directions of the branch.
- Have 2^n two bit saturating counters per branch, one for each possible history pattern (it'll be in $\{0, 1\}^n$).

Two Level Local Diagram



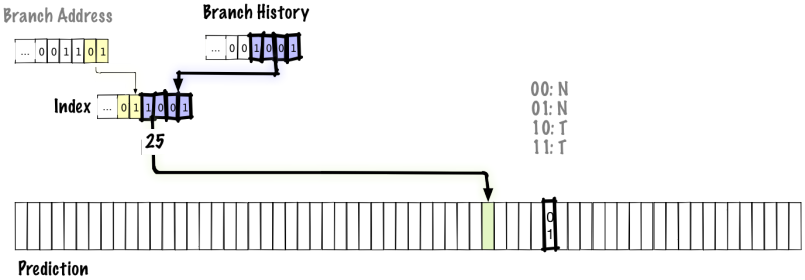
[2]

[2]Image credit: `https://danluu.com`

Local vs. Global Prediction

- Local: maintain a history for each individual branch instruction.
- Global: use a *global* history table instead of one local to each branch.
  - If branch directions are correlated with each other, performs better, else caused contamination between branches.

Two Level Global Diagram (gselect)



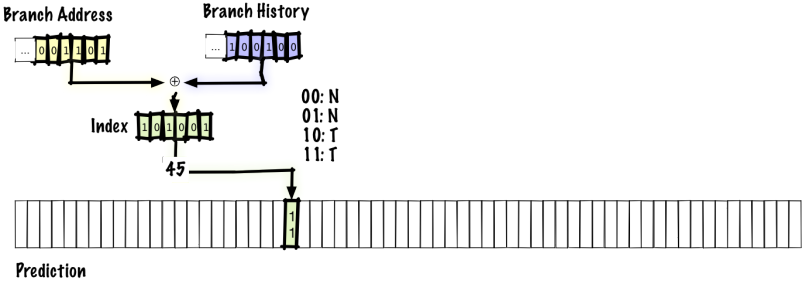
[3]

[3]Image credit: https://danluu.com

gshare

- `gselect`, for a fixed-size prediction table, makes you choose between the number of branch PC bits and the number of global history bits.
- Want both: low interference between branches, and long histories to catch complex patterns.
- To get both, use `gshare`: hash them together with an XOR.

gshare Diagram



Hybridization (McFarling predictors)

- Have multiple prediction mechanisms.
- Pick the predictor that's made the best predictions in the past. (meta-prediction)
- Or use a majority function on predictor outputs.

Lots of other schemes...

- Many schemes attempt to reduce destructive interference due to branch aliasing.
- Others do more involved things.
- bi-mode, gskew, YAGS, TAGE...
- Today, we'll talk about a perceptron-based prediction mechanism.

Key Insight: Branch Prediction is a Boolean Function

- The paper shows that perceptrons can be successfully used to learn certain branches.
- …with a low hardware overhead.
- Note that a primary limitation of perceptrons is that they can only learn linearly separable functions (i.e., a perceptron couldn't correctly learn the XOR function).

Section 2

Overview of Perceptrons and Related Math

Dot Products

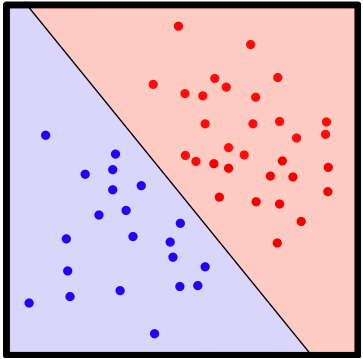
- A vector can be thought of as a list of numbers:[4]
  $\vec{a} = (a_1, a_2, a_3, \ldots, a_n)$ $\vec{b} = (b_1, b_2, b_3, \ldots, b_n)$
- A dot product of two vectors $\vec{a}$ and $\vec{b}$ is defined as:

$$\vec{a} \cdot \vec{b} = \sum_{i=1}^{n} a_i b_i = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$$

[4] Math majors: this is an oversimplification, since this is only true for the special case of coordinate spaces…vectors in general don't need to be "lists".

Linear separability in $\mathbb{R}^2$

Two sets of points (colored say red and blue) on the plane are linearly separable if there exists one line that separates the red points from the blue points.



Linear separability, generalized

- Generalizing to any n dimensional space, sets of points are linearly separable if there is a hyperplane that separates them.
- In the language of convex hulls, two sets of points are linearly separable if their convex hulls are disjoint.

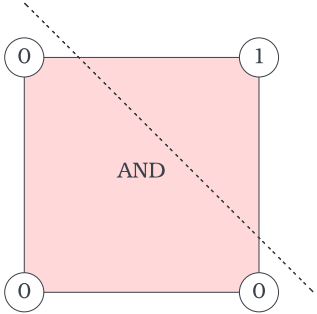
Linear separability, formal definition

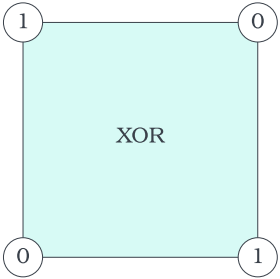
Definition

For sets of points X_0 and X_1 in some n dimensional Euclidean space, X_0 and X_1 are linearly separable if there are $n+1$ real numbers $\vec{w} = (w_i)_{i \in [1,n]}$ and k, such that for all $\vec{x} \in X_0$, $\vec{x} \cdot \vec{w} > k$ and for all $\vec{x} \in X_1$, $\vec{x} \cdot \vec{w} < k$.

Linear separability of Boolean functions

A Boolean function $f : \{0,1\}^n \to \{0,1\}$ is linearly separable if the hypercube of dimension n with vertices assigned to outputs of the function is linearly separable.





Linear Classifiers

- A linear classifier is a function that, given two sets of data and a new data point, decides which set the data point belongs to.
- Given a vector of weights $\vec{w}$ and a feature vector $\vec{x}$, the classifier computes an output score y, given by

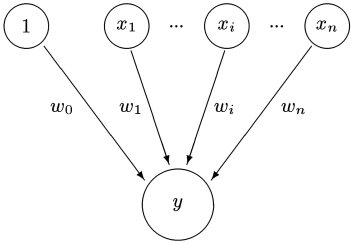
$$y = f(\vec{w} \cdot \vec{x})$$

- Typically f is a threshold function, which is simply a unit step function shifted by θ:

$$f(\vec{w} \cdot \vec{x}) = \begin{cases} 1, & \vec{w} \cdot \vec{x} > \theta \\ 0, & \text{otherwise} \end{cases}$$

Perceptrons

- A perceptron is a *learning* linear classifier.
- It outputs a classification, but also uses the *correct* answer to update the weight vector $\vec{w}$.
- It can be viewed as an artificial neuron whose activation function is the unit step function.



Section 3

The Paper

Citation

D.A. Jimenez and C. Lin, “Dynamic branch prediction with perceptrons,” *Proceedings HPCA Seventh International Symposium on High-Performance Computer Architecture*, Monterrey, Mexico, 2001, pp. 197-206, doi: 10.1109/HPCA.2001.903263.

Background

- When this paper came out (2001), the field was focusing on eliminating aliasing in two level schemes (preventing destructive interference between unrelated branches).
- This paper aimed to improve the accuracy of the prediction itself, by replacing the two-bit saturating counters with perceptrons.

Issue with standard two-level predictors: limited history depth

- The depth of the history buffer D is related to the number of counters C as: $2^D \leq C$ (recall the history pattern acts as an index into the array of counters).
- As we require longer history depths to catch correlated branches with large (temporal) gaps, the counter arrays become prohibitively large.
- Longer history lengths also require longer training times for such schemes.

Why Perceptrons?

- Easy hardware implementation.
- Very easy to understand the decisions they make: just look at the weights vector (in contrast to general neural networks).

The Perceptron Used

- Recall a perceptron's output is given by

$$y = \vec{w} \cdot \vec{x} = \sum_{i=0}^{n} x_i w_i$$

- The paper's perceptron restricts:
  - $w_i \in \mathbb{Z}$ for all w_i – the weight vector,
  - $x_i \in \{-1, 1\}$ for all x_i – the history vector, and
  - $x_0 := 1$
- The second property means that the perceptron input is *bipolar*: either taken or not taken.
- The sign of output y predicts the branch direction.

Training the Perceptron

- Once output y is generated, let $t = \pm 1$ be the actual result of the branch (negative means not taken).
- With threshold θ, here's the training algorithm:

```
if sgn(y) != t or abs(y) <= theta:
    for i in range (0, n + 1):
        w[i] = w[i] + t*x[i]
```

- That is, each `w[i]` is incremented if `x[i]` agreed with the true outcome, and decremented otherwise.
- Positive correlation means a larger weight, negative correlation means a larger negative weight, no/weak correlation means a weight near zero, and contributes less to outputs y.

Overall Algorithm

- Get an index into the array of perceptrons:

```
idx: int = hash(branch_address)
```

- Index into the array and get the perceptron (the weight vector):

```
w: list[int] = perceptron_array[idx]
```

Overall Algorithm

- Take the dot product of the global history register and `w`:

```
y: int = dot_product(w, [1] + global_history)
```

- Make prediction based on sign of `y`:

```
prediction: bool = False if y < 0 else True
```

Overall Algorithm

- Once we have the actual result t, update w:

```
update_weights(w, t)
```

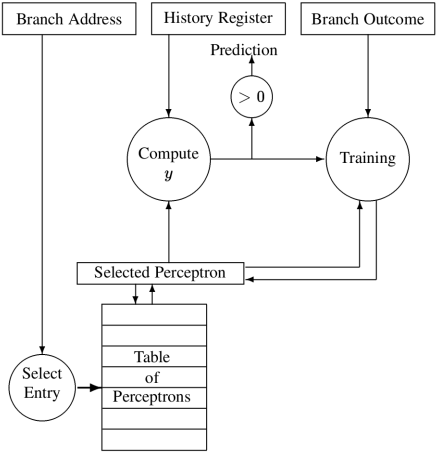
- Write back w to the table:

```
perceptron_array[idx] = w
```

Global v. Local AKA Why the `[1] +`

- The vector $\vec{x}$ is a vector of length $n+1$ that holds the global history in `x[1:]`.
- Note `x[0] == 1` is always true, so `w[0]` learns the bias of that specific branch independent of the global history.

Block Diagram



Microarchitecture: Dot Product

- The real "computation" bits are the dot product and the "training" update to the weights (9 bits).
- Dot product:
  - Don't need multiplication, since the history vector elements are all in $\{-1, 1\}$.
  - Just add `w[i]` if `x[1]` is 1, else subtract. This is very similar to how a digital multiplier works.
  - Paper estimates delay using a 54×54 bit multiplier with a 0.25μm process, concludes this operation can be done in 2 cycles of a 700MHz clock.
  - Also, we only need the sign bit to make the prediction, so others bits can be computed more slowly.

Microarchitecture: Training

- Training can be done bit-parallely since we increment `w[i]` if `x[i]` was right, else decrement.
- The paper sets each weight to 9 bits, so the increment/decrement is also quick.

Parameter Space

Assuming a fixed hardware budget,

- History length: greater length means better prediction but more aliasing since each table entry is larger.
- Representation of weights: could use FP, but signed integers are good enough here.
- Threshold θ: This determines the largest possible output y, hence the largest possible weight w_i, and therefore the "size" of each perceptron table entry.

Results

- This paper is very influential: lots of other papers are based on it, and perceptron-based branch predictors are used in Samsung's M1, AMD's Zen (and likely other AMD chips).
- Still, looking at methodology and experimental results is valuable.

Compared Against

- `gshare`
- `bimode`: A kind of branch predictor designed to prevent destructive aliasing as much as possible.
- Show that a hybrid of `gshare` and the perceptron predictor has good performance since they have complementary strengths.

Instrumentation

- Compiled programs w/gcc, ran on AMD K6-III under Linux.
- Instrumented the binaries so every branch called a trace-generation function.
- Took traces with information about branches and their directions and fed them to a branch predictor simulator.

Benchmarks and Tuning

- SPEC 2000 integer benchmarks.
- ~100 million branch traces, equivalent to simulating roughly 0.5 billion total instructions.
- Tuned all predictors by searching the parameter spaces for optimal values for various hardware budgets.

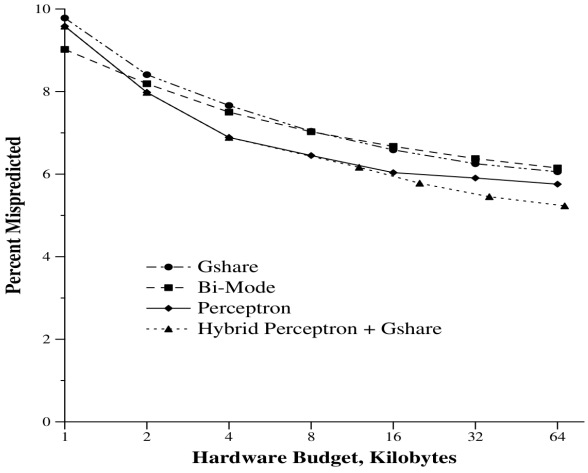
Optimal History Lengths

Hardware budget in kilobytes	History Length		
	gshare	bi-mode	perceptron
1	6	7	12
2	8	9	22
4	8	11	28
8	11	13	34
16	14	14	36
32	15	15	59
64	15	16	59
128	16	17	62
256	17	17	62
512	18	19	62

On History Lengths

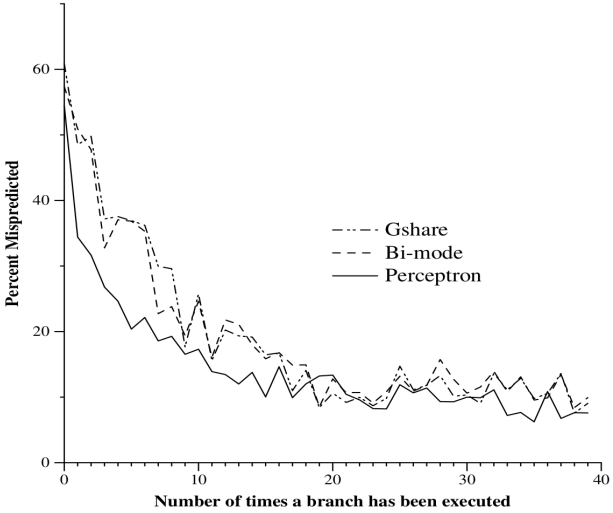
- Schemes like `gshare`'s table sizes scale exponentially (as 2^N) with the length of the history buffer.
- The perceptron predictor scales *linearly* — if you increase the size of your history buffer by 1, each perceptron needs 1 extra bit.
- This allows the perceptron predictor to look at far longer histories for the same hardware budget.

Hardware Budget v. Misprediction Rate



Perceptron vs. other techniqes, Harmonic Mean

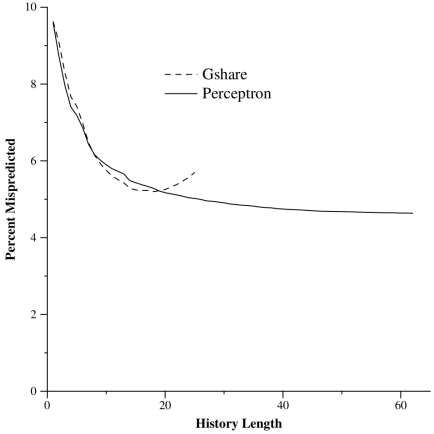
Training Times: Perceptrons Learn Faster



Why Do Perceptrons Do Well?

- Main advantage is being able to use longer history lengths.
- When `gshare` and perceptron use histories of the same length, `gshare` outperforms perceptron due to:
  - Destructive aliasing in perceptron
  - `gshare` can learn non linearly separable Boolean functions.

History Length v. Misprediction



When Does it Do Well

- For n bits of history, a static branch B, and a history vector h_n, let $f_B(h_n)$ be the Boolean function that best predicts B.
- If f_B is not linearly separable, then `gshare` will usually outperform the perceptron.
- …unless the branch requires long histories to be accurately predicted. Then, the perceptron predictor still does better.

Additional Advantages

- Output y is also a confidence estimator — if the confidence is low, the pipeline can be gated (as discussed in a SP23 meeting).
- The weight vector of the perceptron once trained can give data to other modules about branch correlation.

That's All

Thanks for coming!